

# Primordial Black Holes from the collapse of Fermi-balls\*

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This project involves the study of Primordial Black Hole (PBH) production through the collapse of Fermi-balls formed during a first-order phase transition in the early universe. I use computational and numerical methods to study the shape of the potential that triggers this phase transition and analyze the profile of PBHs formed to see if they can explain the current dark matter energy density.

## I. BACKGROUND AND THEORY

Primordial Black Holes (PBHs), formed early in the universe's history, remain an interesting dark matter candidate. While they evade direct detection, observational searches have been able to set constraints on their mass spectrum and abundance. Figure 1 shows that PBHs in the asteroid-mass range ( $\sim 10^{17}$ g to  $10^{22}$ g) have not been constrained, and may have the abundance necessary to explain all of dark matter.

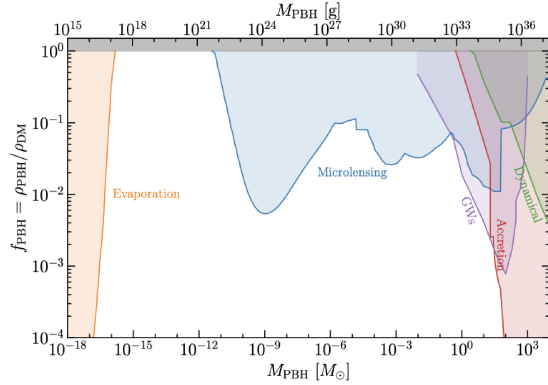


FIG. 1: Constraints on the fraction of DM in the form of PBHs (Green, 2024)

The standard PBH-forming mechanism is through the collapse of overdensities in the early universe. Simply put, as the horizon expands during the radiation-domination era, density perturbations created during inflation enter the horizon. Some of these perturbations may have been large enough to collapse gravitationally into black holes upon entering the horizon.

In this project, I will explore a second mode of PBH formation: the collapse of Fermi-balls formed

during a first-order phase transition in the early universe.

This mode of PBH formation requires a scalar field  $\phi$  coupled to a Dirac fermion  $\chi$  in the early universe. This system obeys the Lagrangian:

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - U(\phi, T) + \bar{\chi}i\not{\partial}\chi + y\phi\bar{\chi}\chi \quad (1)$$

The terms, in order, are the kinetic term for the scalar field, the potential for the scalar field, the kinetic term for the spinor field, and the scalar-spinor interaction term with Yukawa coupling  $y$ .

The scalar potential  $U(\phi, T)$  has two degenerate vacua at  $\langle\phi\rangle = 0$  and  $w$  at a critical temperature  $T_c$ . As the universe cools, the minimum at  $w$  loses energy, and is now called the true vacuum. The scalar field, originally in the false vacuum  $\langle\phi\rangle = 0$  tunnels to the true vacuum, executing a first-order phase transition. This transition takes place through true vacuum bubbles nucleating in the false vacuum, growing, and subsequently connecting with each other, causing the false vacuum to shrink.

The fermions, which have a mass of  $y\langle\phi\rangle = 0$  in the false vacuum, gain a large mass  $M_\chi = yw$  in the true vacuum. Assuming that the fermions do not have sufficient thermal kinetic energy, gaining this mass would violate energy conservation, and so they remain trapped in the false vacuum and cannot penetrate into the true vacuum.

As the false vacuum shrinks, the fermions and anti-fermions annihilate each other, which, assuming a number density asymmetry  $\eta_\chi$ , leaves behind a number of fermions. These fermions become compressed into non-topological solitons known as Fermi-balls, with the degeneracy pressure balancing the vacuum pressure.

As the universe continues cooling, the Yukawa potential inside these stable Fermi-balls grows, dominating over degeneracy pressure, causing a further collapse into PBHs.

My aim with this project is to analyze and optimize the scalar potential and other Lagrangian parameters to model the production of PBHs, and ascertain the best parameters to produce these PBHs

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at the right abundance and mass to explain dark matter.

In the next section, I will discuss the methodology and numerical techniques I used to accomplish this.

## II. METHODOLOGY

At its core, this simulation involves scanning over a parameter space and computing observables like the mass and density of PBHs formed by each configuration. The pipeline consists of the following steps.

### A. Toy Potential

For this project, I consider a toy potential for the scalar field, defined through a function called `toy_potential` as:

$$U(\phi, T) = \frac{1}{2}(\mu^2 + cT^2)\phi^2 + \frac{\mu_3}{3}\phi^3 + \frac{\lambda}{4}\phi^4 \quad (2)$$

The first term behaves like a mass term for the scalar field, with the second and third terms being cubic and quartic interactions.

For my free parameters, I choose the scalar cubic interaction strength  $\mu_3$  and the Yukawa interaction strength  $y$ . The other parameters in the potential can be written in terms of  $\mu_3$  and  $M_\phi$ , the scalar mass in the true vacuum, as shown in Hong et al.

$$\mu^2 = -\frac{1}{2}(M_\phi^2 + \mu_3 w_0), \quad \lambda = \frac{1}{2w_0} \left( \frac{M_\phi^2}{w_0} - \mu_3 \right) \quad (3)$$

where  $w_0$  is the true vacuum vev at  $T = 0$ .

Thus, my problem becomes: what are the values of  $\mu_3$  and  $y$  that produce PBHs of the right mass and abundance?

First, I scan over ten values of  $\mu_3$  from  $-50$  to  $-10$  GeV, allowing me to test a variety of scalar potentials. For each potential, I scan over  $n_T = 300$  temperature ( $T$ ) values from  $10$  to  $500$  GeV, which acts as my time parameter, allowing me to see the time evolution of each potential. For each temperature, I scan over  $80$  intervals of  $\phi$  in the range  $[-800, 800]$  and use SciPy's `minimize_scalar` function to find the minimum in each interval. This ensures that the function finds local minima and not just the global minimum. The program then deduplicates the minima by considering two minima closer than  $10^{-2}$  in  $\phi$  as a single minimum.

I define  $\phi = 0$  as the false vacuum, and then select the lowest nontrivial minimum as the true vacuum. Finally, the program ensures that it has found

a minimum and not a maximum by using the finite difference approximation for the second derivative, and checking that it is greater than zero.

By doing this over a range of temperatures, I find the temperature that minimizes the quantity  $|\Delta U| = U(0, T) - U(w, T)$ . This is the critical temperature  $T_c$ . The work in this section is done by two functions: `find_minima`, which finds and deduplicates the minima for each temperature, and `find_critical_temperature_scan`, which finds  $T_c$  for a particular  $\mu_3$ .

### B. Simulation

My program then runs through the steps of Fermi-ball and PBH formation. I define a class called `Simulator`, consisting of a number of functions which calculate the necessary quantities and observables. Since the formulas and their explanation require significant detours into theory, I have simply explained the use of each function. The equations and parameters are taken from and motivated by Hong et al (2020) and Kawana and Xie (2021). They are as follows:

1. `nucleation_temperature`: Approximates  $T_n$ , the temperature at which true vacuum bubbles begin to nucleate within the false vacuum as the field tunnels to the true minimum. Rather than calculate this rigorously, I parametrize this as  $0.9T_c$ , slightly lower than the critical temperature, allowing the universe to cool and the true vacuum to drop below the false vacuum in energy.
2. `fermi_ball_number_density`: Estimates the number density  $n_{\text{FB}}$  of Fermi-balls formed.
3. `fermi_ball_charge`: Calculates the charge  $Q_{\text{FB}}$  of the Fermi-balls, using  $n_{\text{FB}}$ .
4. `fermi_ball_mass_radius`: Calculates the mass  $M_{\text{FB}}$  and the radius  $R_{\text{FB}}$  of the Fermi-balls, using  $Q_{\text{FB}}$ .
5. `yukawa_collapse_temperature`: Calculates the temperature at which the Yukawa potential becomes significant, causing the Fermi-balls to collapse into PBHs, using  $R_{\text{FB}}$  and  $Q_{\text{FB}}$ .
6. `pbh_mass`: Calculates the mass of the PBHs formed — approximately similar to the mass of the Fermi-balls, taken as  $0.8M_{\text{FB}}$ .
7. `pbh_abundance`: Computes the energy density  $\rho_{\text{PBH}}$  of PBHs formed by multiplying their number density (same as the Fermi-balls) with their mass.

8. `dark_matter_fraction`: Calculates  $f_{PBH} = \frac{\rho_{PBH}}{\rho_{DM}}$  the fraction of PBHs that make up dark matter in this configuration, i.e., the contribution of PBHs to the present day dark matter abundance.

### C. Best Configuration

I define a function called `run_simulation`, which scans over the parameter space in  $(\mu_3, y)$ , runs through the toy potential and simulation pipeline, and returns a dictionary of results, consisting of the parameter space values as well as the corresponding values of  $M_{PBH}$ ,  $f_{PBH}$ ,  $M_{FB}$ ,  $R_{FB}$ , and  $T_c$ .

After computing the PBH profile for each configuration in the parameter space, a function called `extract_best_configuration` chooses the point in parameter space closest to  $f_{PBH} = 1$  (PBHs explain all of dark matter) and  $M_{PBH} \approx 10^{17}g$  (asteroid-mass region), after excluding configurations that result in infinite or negative PBH masses or infinite abundances. It does this by minimizing the combined loss function:

$$\chi^2 = \left[ \log_{10} \frac{M_{PBH}}{M_{\odot}} - \log_{10} M_{\text{target}} \right]^2 + (f_{PBH} - 1)^2 \quad (4)$$

### D. Monte Carlo Sampling

Once the best configuration is found, a function called `gaussian_sampling` refines it by Gaussian sampling 2000 points in parameter space around this configuration. The previous functions essentially scan parameter space in low resolution, while this function uses Monte Carlo sampling to finely probe the region of interest in parameter space.

It must be noted that instead of sampling in  $(\mu_3, y)$  space, I instead sampled in  $(\log_{10} T_c, \log_{10} y)$  space, keeping  $\mu_3$  fixed, and run through the whole Simulation and Best Configuration pipeline. If I was to sample in  $(\mu_3, y)$  space, I would have to run through the whole Toy Potential pipeline, which, for 2000 samples, is incredibly computationally expensive. Instead,  $T_c$  is used as a proxy for  $\mu_3$  for this section.

### E. Plotting

The function `plot_toy_potential` plots the chosen potential at the best value of  $\mu_3$  for different  $T$  values. `plot_grid_results` then show how a variety

of observables, including  $M_{PBH}$ ,  $M_{FB}$ ,  $R_{FB}$ , scale with the free parameters and with each other. The function also plots the PBH mass spectrum, all configurations in parameter space, and the Monte Carlo refining of the coarse grid. These plots are shown in the Results section.

### F. Main Driver

The main driver of the program defines the parameter space —  $\mu_3 \in [-50, -10]$  (10 values) and  $\log_{10} y \in [-1, 1]$  (12 values) — and calls on the functions I have described in the previous sections in order to run through the entire pipeline. All remaining parameters are fixed to the values in Table 1 as motivated by Hong et al. (2020) and Kawana & Xie (2021).

| Parameter   | Description                     | Value      |
|-------------|---------------------------------|------------|
| $\eta_\chi$ | Fermion–antifermion asymmetry   | $10^{-25}$ |
| $w_0$       | $T = 0$ true vacuum vev         | 400 GeV    |
| $M_\phi$    | Scalar mass in true vacuum      | 100 GeV    |
| $c_{th}$    | Thermal mass coefficient        | 0.4        |
| $g_*$       | Relativistic degrees of freedom | 100        |

TABLE I

## III. RESULTS

Figures 2 to 8 show the results of the program. The best configuration in parameter space was found to be at  $\mu_3 = -50$  and  $y \approx 0.23$ , resulting in the values  $T_c \approx 47.69 \text{ GeV}$ ,  $M_{PBH} \approx 3.75 \times 10^{-33} M_{\odot}$  ( $\sim 7.5g$ ), and  $f_{PBH} \approx 1.89$ . This shows us that, with the best configuration in our chosen parameter space, we are producing PBHs that are far too light compared to the  $\sim 10^{-13}$  asteroid-mass region, even though we are producing them close to the right abundance.

Even with the Monte Carlo refining, the best configuration still gives us PBHs that are  $\mathcal{O}(10^{-33} M_{\odot})$  or  $\mathcal{O}(1g)$ . The refined best PBH dark matter fraction is now  $f_{PBH} \approx 0.85$ .

The best configuration at  $\mu_3 = -50$ , on the edge of the grid, seems to indicate that the ideal point is outside the explored region. I ran the simulation again with two different ranges of  $\mu_3$ ,  $[-200, -50]$  and  $[-1000, -200]$  to see if this results in a better PBH mass spectrum. The best configuration still remains at the edge of the grid, which means that the optimum is well outside the explored region, and the PBH mass from this configuration does

not change significantly from our original parameter space. Since  $M_{PBH}$  depends on  $M_{FB}$ , which is related to the false vacuum-true vacuum energy difference  $U_0$  by  $M_{FB} \sim U_0^{1/4}$ . As  $\mu_3$  changes,  $U_0$  also changes, but due to the  $1/4$  scaling power,  $M_{FB}$  (and thus  $M_{PBH}$ ) changes very slowly.

As another sanity check, I re-ran the program, changing the number  $n_T$  of temperature values I scan over per potential to find  $T_c$  from 300 to 1000, to check the sensitivity of the results to this. The results, and in particular, the PBH mass spectrum, do not change significantly.

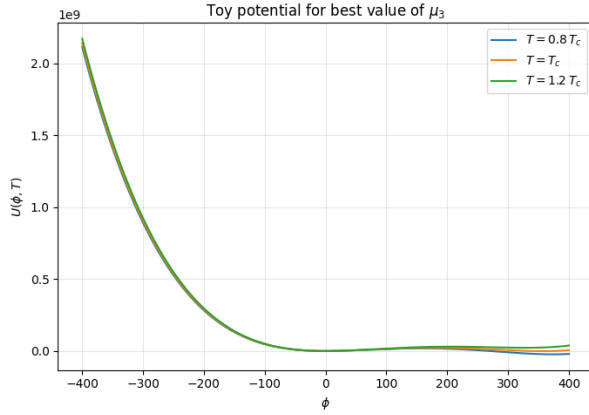


FIG. 2: The temperature evolution of the toy potential at the best value of  $\mu_3$ , with the false vacuum at  $\phi = 0$ . At  $T = T_c$ , the two minima are degenerate, and increasing or decreasing the temperature makes the true vacuum rise above or fall below the false vacuum.

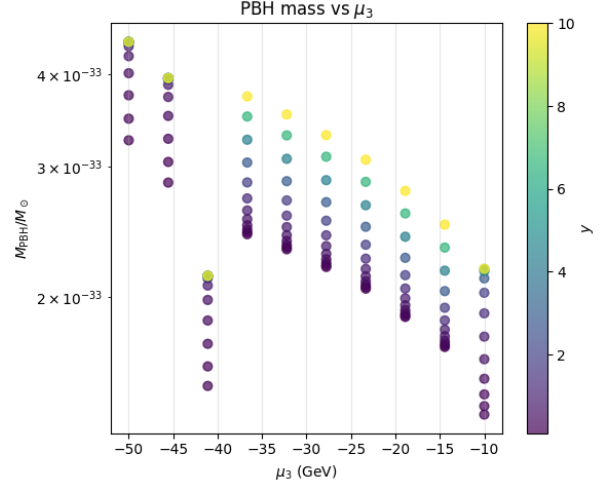


FIG. 3: The mass of PBHs formed in solar mass units against the tested values of  $\mu_3$ . Each point represents a valid PBH forming configuration, and the color bar shows the value of  $y$  for each point.

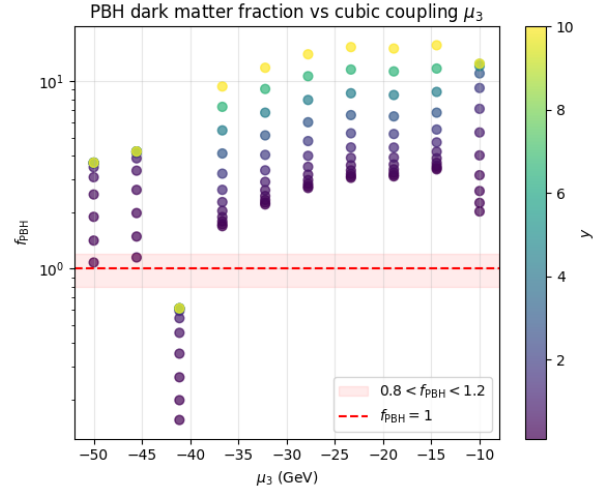


FIG. 4: The PBH contribution to the dark matter energy density against the tested values of  $\mu_3$ .

Each point represents a valid PBH forming configuration, and the color bar shows the value of  $\mu_3$  for each point. Points in the red region have the right energy density to explain dark matter.

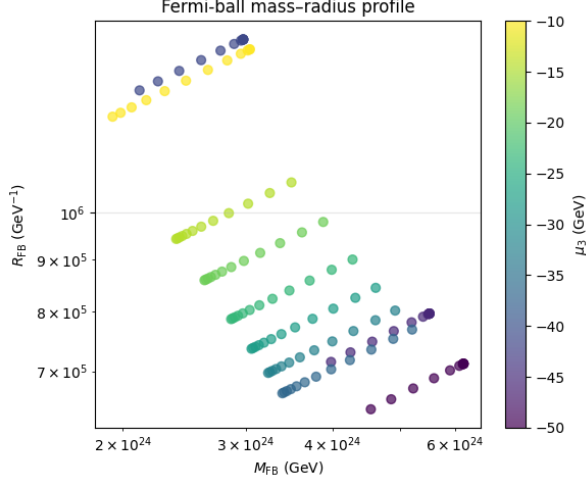


FIG. 5: The Fermi-ball profile, showing the relation between radius and mass at various values of  $\mu_3$ . Each point represents a valid Fermi-ball configuration.

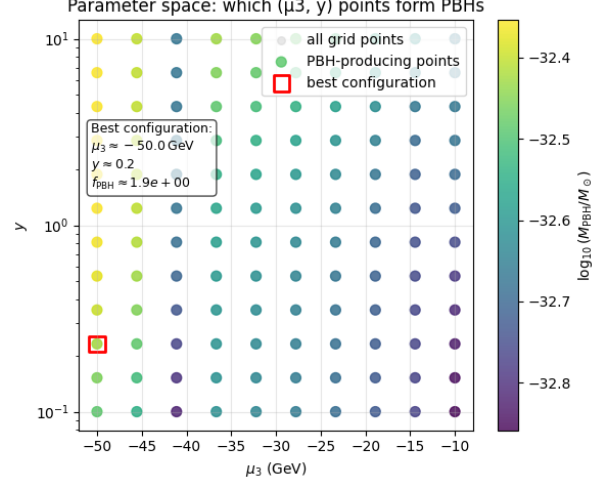


FIG. 7: The entire parameter space in  $(\mu_3, y)$ , with the best configuration marked with a red box.

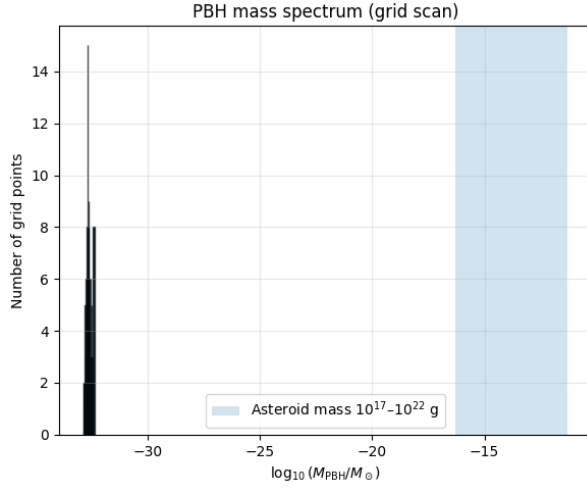


FIG. 6: The PBH mass spectrum shows how the parameter space produces PBHs that are too light to fit into the unconstrained asteroid-mass region shown in blue.

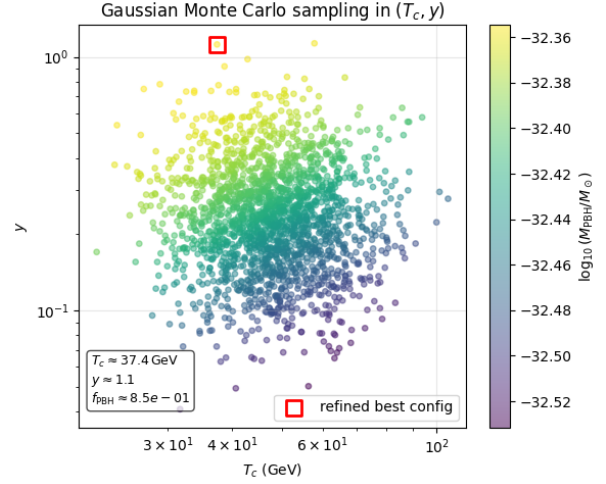


FIG. 8: Gaussian Monte Carlo sampling around the best configuration in the new parameter space  $(T_c, y)$ , with the refined best configuration marked in a red box. Note that the choice of considering  $\mu_3$  as the primary parameter indicates the goal of exploring the shape of the potential which results in a particular PBH spectrum.

#### IV. LINK TO CODE

[Click here](#)

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- [1] A. M. Green, Primordial black holes as a dark matter candidate – a brief overview (2024), arXiv:2402.15211 [astro-ph.CO].
  - [2] J.-P. Hong, S. Jung, and K.-P. Xie, Fermi-ball dark matter from a first-order phase transition, Physical Review D **102**, 10.1103/physrevd.102.075028 (2020).
  - [3] K. Kawana and K.-P. Xie, Primordial black holes from a cosmic phase transition: The collapse of fermi-balls, Physics Letters B **824**, 136791 (2022).